WHITE PAPER · YCM TV-188 B × INDUSTRIAL & GENERAL MANUFACTURING

YCM TV-188 B for Industrial & General Manufacturing

Large-envelope CNC vertical mill for oversized workpieces, complex pocketing, and multi-face machining. This paper covers what the machine does, the industrial & general manufacturing parts we produce with it, the alloys we run, the tolerances and finishes industrial & general manufacturing buyers expect, and the documentation package.

32" W × 70" L × 32" H workpiece envelope
ENVELOPE

±0.0005" routine on critical features
TOLERANCE

1994
SINCE

Same-week / rig-down direct
RESPONSE

Loco Motive · 32" W × 70" L × 32" H workpiece envelope · large-envelope CNC vertical machining center

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Executive summary

Machine: YCM TV-188 B (shop nickname: *Loco Motive*). large-envelope CNC vertical machining center at B&R Productions, running from our New Waverly, Texas shop.

Application: Large-envelope CNC vertical mill for oversized workpieces, complex pocketing, and multi-face machining for Industrial & General Manufacturing customers — Industrial equipment OEMs, custom machinery builders, pump and valve manufacturers outside oilfield, food and pharma equipment suppliers, mining and heavy-equipment operators, reverse-engineering customers, sustainment for out-of-production industrial equipment.

Envelope: 32" W × 70" L × 32" H workpiece envelope. **Tolerance:** ±0.0005" routine on critical features. **Traceability:** Material by heat and lot on every job.

The machine, in context

Loco Motive is where the oversized work lives — valve bodies, wellhead spool adapters, large manifold blocks, and any part that busts the 20-inch class of mills. YCM's TV-188 platform is a heavier, more rigid class of vertical machining center; the extra column and table mass matters when you're taking heavy roughing cuts in Inconel or hogging out a wellhead block.

Why Loco Motive for Industrial work

Large industrial prismatic work — machine bases, large housings, weldment machining, big plates and blocks — needs table capacity as much as travel. 4,409 lb on a 78.7" table with 32" × 70" × 32" of envelope covers the class of part most job-shop verticals can't hold, let alone cut.

What this looks like on a real part

Machining a large fabricated weldment is a different problem from machining a solid block, because the weldment arrives with residual stress locked in from welding. Take a heavy cut off one face and the part moves — sometimes visibly. The strategy is to rough all faces first and let the part relieve, then come back and finish, rather than finishing each face as you go. Finishing face one before roughing face two is how you produce a part that was flat when you measured it and isn't now.

> **When this is the wrong machine**
>
> Round parts belong on a lathe. Smaller industrial brackets and plates are more economical on Pistol Whiskers.

Full specifications

Model	YCM TV-188 B ("Loco Motive")
Type	3-axis CNC vertical machining center — heavy-duty class
Workpiece envelope	32" W × 70" L × 32" H (B&R working spec)
Axis travels	70.9" X × 33.9" Y × 29.5" Z
Table size	78.7" × 33.9" (2,000 × 860 mm)
Max workpiece weight	4,409 lb (2,000 kg)
Spindle taper	BT50
Spindle speed	6,000 RPM standard (10,000 RPM optional)
Spindle power	25 HP standard (30 HP optional), 30-min rating
Automatic tool changer	24 tools standard (40 optional)
Rapid traverse	591 IPM; cutting feed 197 IPM
Control	Fanuc MXP-200i
Structure	Patented T-base for rigidity + chip disposal
Tolerance capability	±0.0005" routine on critical features
Materials	Stainless, carbon steel, tool steel, Inconel, Hastelloy, aluminum, titanium

Values marked "platform" reflect the machine manufacturer's published spec range for this platform family; B&R's working spec is what we quote on this specific unit.

Who this serves in Industrial & General Manufacturing

Typical buyer: Industrial equipment OEMs, custom machinery builders, pump and valve manufacturers outside oilfield, food and pharma equipment suppliers, mining and heavy-equipment operators, reverse-engineering customers, sustainment for out-of-production industrial equipment.

What's hard about industrial & general manufacturing work: Legacy parts with no OEM support, one-off replacement components, small-lot production runs, materials that don't fit standard shop capabilities, and tight lead times on repair work where a down-machine is losing money by the day.

Typical Industrial & General Manufacturing parts we produce on Loco Motive

- Large industrial equipment housings and gearboxes
- Extended pump bodies and machinery frames
- Large manifold and valve bodies for industrial applications
- Machining fixtures and tooling for industrial customers
- Large reverse-engineered legacy replacement parts

Above list is representative; if your part isn't shown, call and ask — most oilfield / aerospace / defense / industrial turned or milled work fits somewhere in the shop.

Alloys we run for Industrial & General Manufacturing

Standard range: Stainless (304, 316, 17-4 PH) · Tool steels · 4140 / 4340 · Brass · Bronze · Aluminum · Cast iron · Specialty alloys per customer requirement.

Industrial work covers the widest alloy range because the applications are the widest. Food and pharma often specify 316L or 17-4 PH; mining and heavy equipment specify tool steels and 4140/4340; pump manufacturers outside oilfield use everything from bronze bushings to stainless impellers. We work with the customer's spec sheet — if you tell us the alloy, we'll tell you honestly whether we can machine it and how quickly.

> **What "we run these weekly" means:** we stock or reliably source the common alloys in the industry's typical spec range. Sharp tooling, proven feeds and speeds, and process control specific to each alloy family. Your part is not the one we're experimenting on.

How we machine it — our 5-step process

This is the process B&R runs on every job, industrial & general manufacturing or otherwise. The specifics of feeds, speeds, and tooling shift by alloy and machine; the discipline is universal.

1. **Material verification.** Every job starts with material traceability. Heat and lot numbers documented against the mill test report. For customer-supplied material, we verify against the CofC before touching the stock.

2. **First-article layout.** Stock is measured and laid out before any cutting starts. Concentricity, roundness, and squareness of the starting material established; if the stock is out of tolerance we call the customer before wasting time on a bad blank.

3. **Roughing with process control for the alloy.** Feeds, speeds, and coolant strategy set for the specific alloy family — sharp coated carbide (or PCD for high-volume Ti Grade 5), positive rake, high-pressure through-tool coolant on 2507 and Inconel to control cutting temperature. Interrupted cuts get extra attention because they cycle heat into the workpiece.

4. **Finish pass to spec.** Finish pass with a fresh insert, controlled DOC, and measured surface-finish targets. Critical features (bore concentricity, seat grooves, sealing surfaces) are held tighter than most job-shop lathes can offer.

5. **CMM verification + documentation.** Every critical dimension CMM-verified. First-article report generated. Certificate of conformance issued with delivery. If your quality group needs a documentation package beyond that, tell us with the RFQ and we'll be straight about what we can and can't produce in-house.

Tolerance and finish expectations in Industrial & General Manufacturing

±0.0005" routine on critical features. Industrial applications range from precision (pump seal areas, close-fit bushings) to structural (mining and heavy-equipment brackets); we adjust the inspection plan to what the part actually needs. Surface finish measured on parts where finish drives function.

Documentation and quality workflow

Documentation calibrated to customer requirement. Material traceability standard for critical parts. Certificates of conformance issued with delivery. NDAs on request for proprietary designs or reverse-engineered legacy parts.

Response time and emergency work

Down-machine and production-critical work prioritized. Lead times honestly quoted — no false promises. For legacy reverse-engineering, we typically confirm feasibility on the first phone call. Call (936) 291-7827.

How we compare

Compared to sourcing a new OEM part: we're often faster and cheaper for one-offs and small runs, and for legacy parts where the OEM has moved on we're often the only realistic option. Compared to a cheapest-bidder job shop: we hold tolerance, document material, and don't ruin the part.

Working with B&R Productions

New customer or existing, the process starts the same way: call (936) 291-7827 or send a print through the quote form. Prints are accepted as PDF, STEP, IGES, DWG, or DXF; if you only have a sample part, we reverse-engineer routinely. NDAs on request, and prints stay in-shop rather than going out to third-party vendors. Note for defense and aerospace buyers: B&R is not ITAR- or AS9100-registered, so export-controlled prints and AS9100 flow-down programs need a registered shop — we'll tell you that on the first call, not after the PO.

Quotes typically come back within 24 hours for standard work, same-day for repeat customers on stocked materials, and near-immediately by phone for emergency and rig-down situations. Lead times are quoted honestly — we don't promise Wednesday and deliver next month.

Frequently asked questions

8 questions covering the YCM TV-188 B platform and Industrial & General Manufacturing work. Also indexed as FAQ schema for AI answer engines.

What's the max part size on the YCM TV-188B?

B&R's stated working envelope is 32" W × 70" L × 32" H. The YCM TV-188B platform has a 78.7" × 33.9" table, travels of 70.9" X × 33.9" Y × 29.5" Z, and max workpiece weight of 4,409 lb (2,000 kg).

Does the TV-188B have a high-speed spindle option?

Standard spindle is 6,000 RPM with a 10,000 RPM option on some builds. BT50 tooling, 25 HP standard (30 HP optional, 30-min rating).

What alloys can the TV-188B run at production speeds?

Full oilfield + aerospace alloy range including Inconel and Hastelloy. The T-base structural design and heavy castings keep it stable through the heavy roughing cuts these alloys demand.

What kind of industrial customers does B&R work with?

Industrial equipment OEMs, custom machinery builders, non-oilfield pump and valve manufacturers, food and pharma equipment suppliers, mining and heavy-equipment operators, and any customer who needs a legacy or reverse-engineered part made right.

Do you machine one-off replacement parts?

Yes — a lot of our industrial work is exactly this. Send the sample or drawing; if the material is on the shelf, we're often faster than the OEM. Reverse-engineered legacy parts across almost every sector.

What materials do you run for industrial work?

Full range — stainless (304, 316, 17-4 PH), tool steels, 4140/4340, brass, bronze, aluminum, cast iron, and specialty alloys per customer requirement. If you need something unusual, ask.

Do you offer emergency turnaround on down-machine parts?

Yes for repeat customers with material on the shelf. Same-day and next-day realistic when we already have the alloy. Call (936) 291-7827 and describe what's down.

Can you do small-batch production or is B&R just one-offs?

Both. Prototype and small-batch through repeat production. Small production runs where a big shop's minimum order would kill the economics are exactly where we fit.

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Ready to talk about your Industrial & General Manufacturing work?

Send us a print or a sample photo and specs. Emergencies: call direct and say "rig-down."

Published by B&R Productions — a precision CNC machining shop in New Waverly, Texas, in business since 1994. Serving oil & gas, aerospace, defense, and industrial customers across Texas and the Gulf Coast.

Written by the B&R Productions team. Machine specs verified against manufacturer data. Alloy and process notes reflect shop practice as of 2026-01-15.